



AMANUEL DEMIREW

FULL STACK & AI ENTHUSIAST

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intro-steel-sigma.vercel.app | Addis Ababa, Ethiopia

PROFESSIONAL SUMMARY

Full-Stack Engineer specializing in AI systems, backend architecture, automation, and scalable web applications. Crafting elegant solutions to complex problems. Combining over 4 years of hands-on experience in full-stack development with deep expertise in Python, FastAPI, and cutting-edge AI orchestration (LangChain, LlamaIndex, n8n). Proven track record of designing high-throughput API architectures, deploying production-ready agentic workflows, and reducing data processing latency by 25%. Passionate about bridging the gap between advanced machine learning research and scalable, resilient, user-centric web applications.

SKILLS

PROGRAMMING LANGUAGES:	Python Java JavaScript JavaScript TypeScript
FRAMEWORKS & LIBRARIES:	FastAPI Django Next.js Spring Boot LangChain&LlamaIndex
TOOLS & TECHNOLOGIES:	Git GitHub GitLab AWS Docker Linux REST APIs n8n
DATABASES:	PostgreSQL MySQL MongoDB SQLite
OPERATING SYSTEMS:	Windows macOS Linux Distributions Chrome OS

PROFESSIONAL EXPERIENCE

BACKEND LEAD DEVELOPER

Startup Misk Company - Addis Ababa, Ethiopia | 2025 – 2025

- **Leading** the backend development team in architecting and implementing scalable, high-performance server-side solutions.
- **Overseeing** API design and database management systems to ensure exceptional reliability, low latency, and platform scalability.
- **Mentoring** junior developers and establishing engineering best practices for code quality, testing suites, and CI/CD deployment pipelines.
- **Collaborating** closely with cross-functional teams to align backend architecture with strategic business objectives and user needs.

TECHNICAL INTERN

iCog Labs - Addis Ababa, Ethiopia | | October, 2024 – August, 2025

- **Completed** an intensive 11-month tenure focusing on machine learning research, software optimization, and agentic workflows.

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- **Improved** machine learning predictive accuracy by 25% and reduced processing time by 15% through meticulous algorithm optimization.
- **Contributed** as Project Manager for the TaskTraker application, enhancing overall workflow and delivery efficiency by 20%.
- **Designed** and implemented an interactive AI chatbot agent for TaskTraker, significantly increasing user interaction and daily productivity.
- **Analyzed** massive datasets using Python and SQL to enable data-driven decision-making for complex technical operations.

AI & BACKEND ENGINEER (FREELANCE)

Upwork - Remote | | August, 2024 – Present

- **Designed and deployed** production-ready AI agents for task automation, enhancing operational workflows for international clients.
- **Optimized** web applications built with Django, resulting in a 30% increase in application speed and a 20% boost in user engagement.
- **Integrated** modern AI libraries like LangChain and LlamaIndex with FastAPI to build scalable, high-throughput content and backend pipelines.

EDUCATION

FULL STACK SOFTWARE ENGINEERING (BACKEND) & AI CAREER ESSENTIALS

ALX Africa, Online | August, 2024

Graduated with exceptional performance metrics: 124% overall performance in Software Engineering and 134% in Backend Engineering.

BACHELOR OF SCIENCE IN CHEMICAL ENGINEERING

Addis Ababa University, Addis Ababa, Ethiopia | April, 2024

Developed a foundational analytical mindset, applying mathematical modeling, thermodynamics, and complex system designs to modern engineering challenges.

CERTIFICATIONS

Big Data Foundations - Level 1 IBM - Online | Date: August, 2024

Python (Basic) Certificate HackerRank - Online | Date: June, 2024

Introduction to Large Language Models DeepLearning.AI - Online | Date: September, 2024

Agentic Application Course DeepLearning.AI - Online | Date: October, 2024

Unstop Weekly Coding Challenge Certificate Unstop - Online | Date: November, 2024

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PROJECTS

AI-POWERED STUDY ASSISTANT PLATFORM

Personal Project | January, 2025 – April, 2025 Description: Built a comprehensive AI-powered study assistant platform leveraging LangChain, LlamaIndex, FastAPI, Next.js, and n8n automation. The platform transforms dense textbooks and PDFs into highly structured, user-friendly notes, worked examples, and interactive exercises. Engineered robust backend APIs and automated content generation pipelines using Python. Link to the project: <https://github.com/ad-code1993>

TASKTRAKER AI CHATBOT AGENT

iCog Labs | October, 2024 – August, 2025

Description: Designed, built, and deployed an AI chatbot agent integrated into TaskTraker to facilitate interactive task management and increase productivity. Optimized the underlying LLM workflows to handle real-time queries with minimal latency. Link to the project: [View on GitHub](#)

3D PORTFOLIO WEBSITE

Personal Project | January, 2025

Description: Developed an immersive, interactive 3D portfolio web application to visually showcase software engineering work, projects, and skills. Leveraged advanced frontend tools to create responsive 3D animations and environments. Link to the project: [View the Project \(3dportfolio-olive\)](#)